

Model Comparison - Without Sex/Race

Model	Logistic Regression	Difference (Regression)	Random Forest	Difference (Forest)
Train Accuracy	0.848	-0.001	0.914	0
Train Precision	0.739	0	0.904	+0.003
Train Recall	0.603	-0.005	0.734	+0.001
Train F1-Score	0.664	-0.003	0.810	0
Test Accuracy	0.846	-0.001	0.859	+0.001
Test Precision	0.735	+0.001	0.786	+0.002
Test Recall	0.601	-0.005	0.596	+0.001
Test F1-Score	0.661	+0.003	0.678	0

Comparing the results of both models with and without the sex and race columns we can see there are no major differences. The logistic regression decreases in most metrics overall, a slight increase in testing F1 and precision with everything else decreasing slightly. The tree model improves slightly across the board, with both F1 metrics and train accuracy staying the same. In both models, metrics change only a fraction of a percent, this implies that sex and race are not substantially important to either model's classification justification.